Roll No....11035.....

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA

End Semester Exam (Regular & Re-appear)- May/June 2025

Branch- ME, PIE and R&A

Semester: II

Date: 20/05/2025
Time: 3 Hours
Max. Marks:50

Course Name: Engineering Practice
Course Code: MEIC-102

Instructions:

1. Assume suitable data wherever essential and mention it clearly.
2. Writing appropriate units, nomenclature, and **schematics diagram is an integral part of the answer, draw wherever required.**
3. Be precise and don't provide unnecessary information that has not been asked.
4. Attempt all the parts of a question at one place maintaining their sequence.

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1. Explain the key components of a lathe machine with neat schematic diagram and discuss how each contributes to its functionality. Describe threading and taper turning operations that can be performed using a lathe machine. [13 Marks]

2. Explain the working principle of Arc Welding (AW) with neat schematic diagram. Which polarity mode in arc welding is best suited for joining thin metal plates, and why? [10 Marks]

3. Write short notes on the following tools with functionality, role and neat schematic diagram [12 Marks]
   (a) Try square (b) File and its types (c) Saw (b) Jack Plane

4. Draw the schematic diagram of a sand mold in casting, illustrating its various components. Explain the importance of the different parts of the gating system in the mold. [10 Marks]

5. Explain the difference between Soldering and Brazing in tabular form only. [5 Marks]